2.2 Sample Space and Events

Consider an experiment whose outcome is not predictable with certainty. However, although the outcome of the experiment will not be known in advance, let us suppose that the set of all possible outcomes is known. This set of all possible outcomes of an experiment is known as the *sample space* of the experiment and is denoted by *S*. Following are some examples:

1. If the outcome of an experiment consists of the determination of the sex of a newborn child, then
   $S=\{g, b\}$

   where the outcome g means that the child is a girl and b that it is a boy.
2. If the outcome of an experiment is the order of finish in a race among the 7 horses having post positions 1, 2, 3, 4, 5, 6, and 7, then
   $S=\{\text{all } 7! \text{ permutations of } (1, 2, 3, 4, 5, 6, 7)\}$

   The outcome (2, 3, 1, 6, 5, 4, 7) means, for instance, that the number 2 horse comes in first, then the number 3 horse, then the number 1 horse, and so on.
3. If the experiment consists of flipping two coins, then the sample space consists of the following four points:
   $S=\{(h, h), (h, t), (t, h), (t, t)\}$

   The outcome will be (h, h) if both coins are heads, (h, t) if the first coin is heads and the second tails, (t, h) if the first is tails and the second heads, and (t, t) if both coins are tails.
4. If the experiment consists of tossing two dice, then the sample space consists of the 36 points
   $S=\{(i, j): i, j=1, 2, 3, 4, 5, 6\}$

   where the outcome (i,j) is said to occur if i appears on the leftmost die and j on the other die.
5. If the experiment consists of measuring (in hours) the lifetime of a transistor, then the sample space consists of all nonnegative real numbers; that is,
   $S=\{x: 0\leq x<\infty\}$

Any subset E of the sample space is known as an *event*. In other words, an event is a set consisting of possible outcomes of the experiment. If the outcome of the experiment is contained in E, then we say that E has occurred. Following are some examples of events.

In the preceding Example 1, if $E=\{g\}$, then E is the event that the child is a girl. Similarly, if $F=\{b\}$, then F is the event that the child is a boy.

In Example 2, if

$E=\{\text{all outcomes in S starting with a 3}\}$

then E is the event that horse 3 wins the race.

In Example 3, if $E=\{(h, h), (h, t)\}$, then E is the event that a head appears on the first coin.

In Example 4, if $E=\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$, then E is the event that the sum of the dice equals 7.

In Example 5, if $E=\{x: 0\leq x\leq 5\}$, then E is the event that the transistor does not last longer than 5 hours.

For any two events E and F of a sample space S, we define the new event $E\cup F$ to consist of all outcomes that are either in E or in F or in both E and F. That is, the event $E\cup F$ will occur if *either E* or F occurs. For

instance, in Example 1, if E={g} is the event that the child is a girl and F={b} is the event that the child is a boy, then

$E \cup F = \{g, b\}$

is the whole sample space S. In Example 3, if E={(h, h), (h, t)} is the event that the first coin lands heads, and F={(t, h), (h, h)} is the event that the second coin lands heads, then

$E \cup F = \{(h, h), (h, t), (t, h)\}$

is the event that at least one of the coins lands heads and thus will occur provided that both coins do not land tails.

The event $E \cup F$ is called the *union* of the event E and the event F.

Similarly, for any two events E and F, we may also define the new event *EF*, called the *intersection* of E and F, to consist of all outcomes that are both in E and in F. That is, the event *EF* (sometimes written $E \cap F$) will occur only if both E and F occur. For instance, in Example 3, if E={(h, h), (h, t), (t, h)} is the event that at least 1 head occurs and F={(h, t), (t, h), (t, t)} is the event that at least 1 tail occurs, then

$EF = \{(h, t), (t, h)\}$

is the event that exactly 1 head and 1 tail occur. In Example 4, if E={(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)} is the event that the sum of the dice is 7 and F={(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)} is the event that the sum is 6, then the event *EF* does not contain any outcomes and hence could not occur. To give such an event a name, we shall refer to it as the null event and denote it by Ø. (That is, Ø refers to the event consisting of no outcomes.) If EF=Ø, then E and F are said to be *mutually exclusive*.

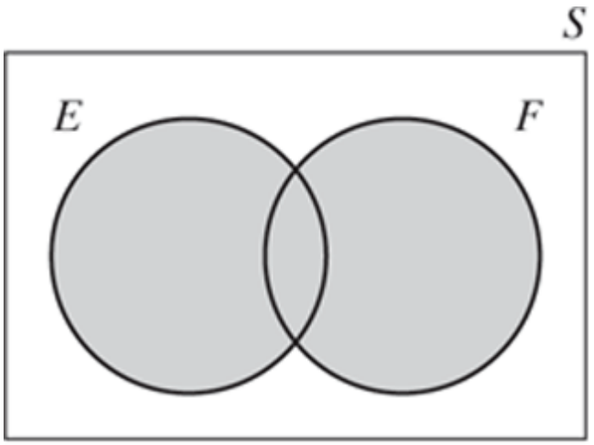
We define unions and intersections of more than two events in a similar manner. If E1, E2, ... are events, then the union of these events, denoted by $\cup_{n=1}^{\infty} E_n$, is defined to be that event that consists of all outcomes that are in En for at least one value of n=1, 2, ... Similarly, the intersection of the events En, denoted by $\cap_{n=1}^{\infty} E_n$, is defined to be the event consisting of those outcomes that are in all of the events En, n=1, 2, ...

Finally, for any event E, we define the new event Ec, referred to as the *complement* of E, to consist of all outcomes in the sample space S that are not in E. That is, Ec will occur if and only if E does not occur. In Example 4, if event E={(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)}, then Ec will occur when the sum of the dice does not equal 7. Note that because the experiment must result in some outcome, it follows that $S^c = \emptyset$.

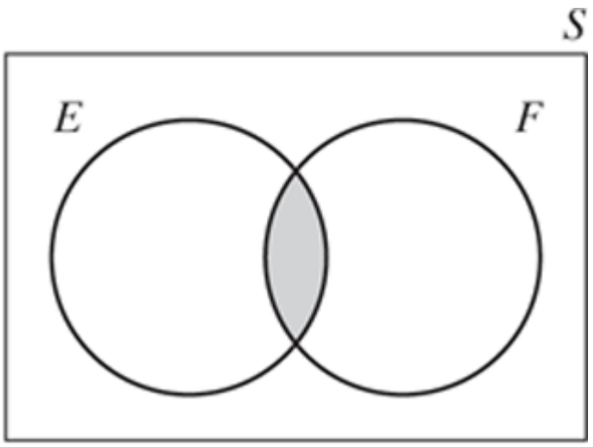
For any two events E and F, if all of the outcomes in E are also in F, then we say that E is *contained* in F, or E is a *subset* of F, and write $E \subset F$ (or equivalently, $F \supset E$, which we sometimes say as F is a *superset* of E). Thus, if $E \subset F$, then the occurrence of E implies the occurrence of F. If $E \subset F$ and $F \subset E$, we say that E and F are equal and write E=F.

A graphical representation that is useful for illustrating logical relations among events is the Venn diagram. The sample space S is represented as consisting of all the outcomes in a large rectangle, and the events E, F, G, ... are represented as consisting of all the outcomes in given circles within the rectangle. Events of interest can then be indicated by shading appropriate regions of the diagram. For instance, in the three Venn diagrams shown in **Figure 2.1**, the shaded areas represent, respectively, the events $E \cup F$, *EF*, and Ec. The Venn diagram in **Figure 2.2** indicates that $E \subset F$.

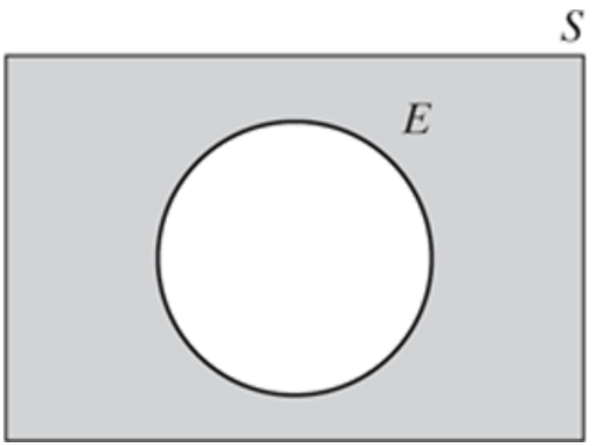
Figure 2.1 Venn diagrams.



(a) Shaded region: E∪F.

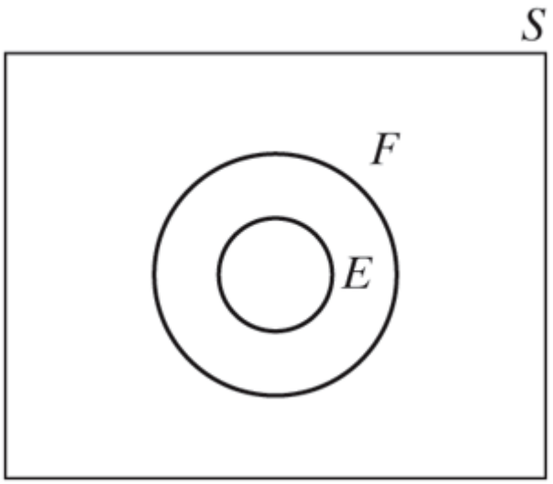


(b) Shaded region: *FF*



(b) Shaded region: E^c

Figure 2.2 E⊂F.

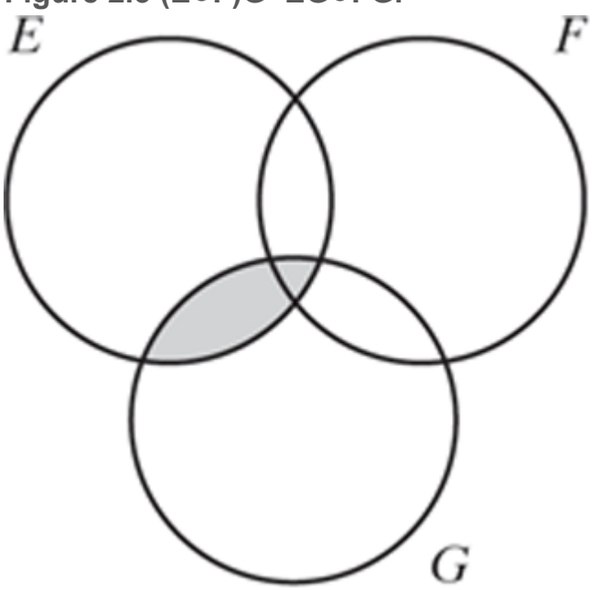


The operations of forming unions, intersections, and complements of events obey certain rules similar to the rules of algebra. We list a few of these rules:

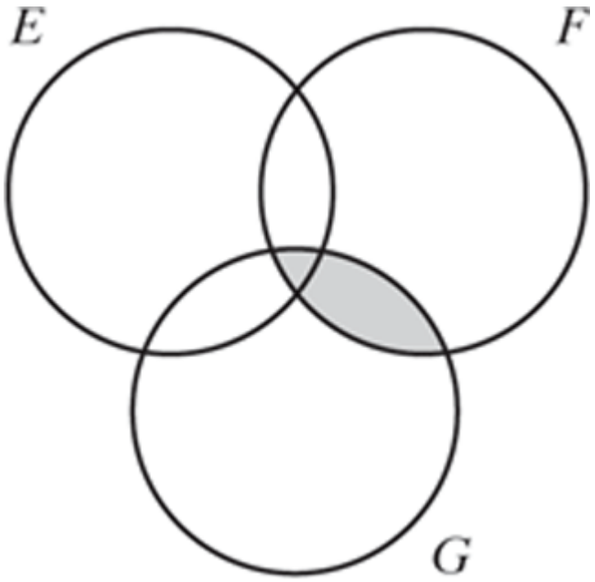
Commutative laws E∪F =F∪E EF =FE Associative laws (E∪F)∪G =E∪(F∪G) (EF)G =E(FG)
Distributive laws (E∪F)G =EG∪FG EF∪G =(E∪G)(F∪G)

These relations are verified by showing that any outcome that is contained in the event on the left side of the equality sign is also contained in the event on the right side, and vice versa. One way of showing this is by means of Venn diagrams. For instance, the distributive law may be verified by the sequence of diagrams in **Figure 2.3**.

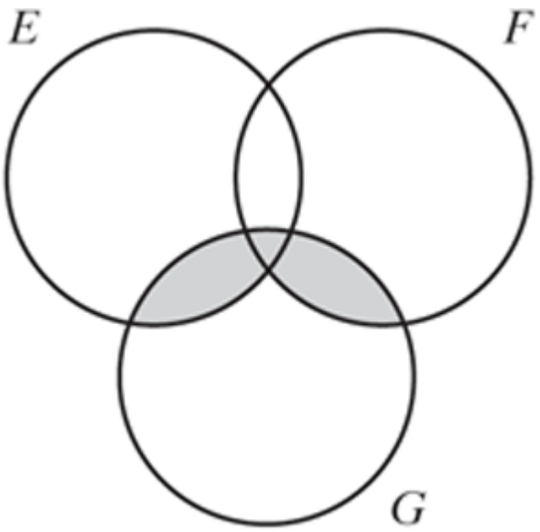
Figure 2.3 (E∪F)G=EG∪FG.



(a) Shaded region: *EG*.



(b) Shaded region: *FG*.



(c) Shaded region: (E∪F)G.

The following useful relationships among the three basic operations of forming unions, intersections, and complements are known as *DeMorgan's laws*:

(∪ i=1 n E i) c = ∩ i=1 n E i c (∩ i=1 n E i) c = ∪ i=1 n E i c

For instance, for two events *E* and *F*, DeMorgan's laws state that

(E∪F)c=EcFc and (EF)c=Ec∪Fc

which can be easily proven by using Venn diagrams (see **Theoretical Exercise 7**).

To prove DeMorgan's laws for general n, suppose first that x is an outcome of (∪i=1nEi)c. Then x is not contained in ∪i=1nEi, which means that *x* is not contained in any of the events Ei, i=1, 2, ... , n, implying that x is contained in Eic for all i=1, 2, . . . , and thus is contained in ∩i=1nEic. To go the other way, suppose that *x* is an outcome of ∩ i=1 n E i c . Then x is contained in Eic for all i=1, 2, . . . , n, which means that x is not contained in Ei for any i=1, 2, . . ., n, implying that *x* is not contained in ∪ i n E i , in turn implying that x

is contained in (∪1nEi)c. This proves the first of DeMorgan’s laws.

To prove the second of DeMorgan’s laws, we use the first law to obtain

(∪ i=1 n E i c) c = ∩ i=1 n (E i c) c

which, since (Ec)c=E, is equivalent to

(∪1nEic)c=∩1nEi

Taking complements of both sides of the preceding equation yields the result we seek, namely,

∪ 1 n E i c = (∩ 1 n E i) c

2.3 Axioms of Probability

One way of defining the probability of an event is in terms of its long run *relative frequency*. Such a definition usually goes as follows: We suppose that an experiment, whose sample space is S, is repeatedly performed under exactly the same conditions. For each event E of the sample space S, we define n(E) to be the number of times in the first n repetitions of the experiment that the event E occurs. Then P(E), the *probability* of the event *E*, is defined as

$$P(E)=\lim_{n\to\infty} n(E)n$$

That is, P(E) is defined as the (limiting) proportion of time that E occurs. It is thus the limiting relative frequency of E.

Although the preceding definition is certainly intuitively pleasing and should always be kept in mind by the reader, it possesses a serious drawback: How do we know that n(E)/n will converge to some constant limiting value that will be the same for each possible sequence of repetitions of the experiment? For example, suppose that the experiment to be repeatedly performed consists of flipping a coin. How do we know that the proportion of heads obtained in the first n flips will converge to some value as n gets large? Also, even if it does converge to some value, how do we know that, if the experiment is repeatedly performed a second time, we shall obtain the same limiting proportion of heads?

Proponents of the relative frequency definition of probability usually answer this objection by stating that the convergence of n(E)/n to a constant limiting value is an assumption, or an *axiom*, of the system. However, to assume that n(E)/n will necessarily converge to some constant value seems to be an extraordinarily complicated assumption. For, although we might indeed hope that such a constant limiting frequency exists, it does not at all seem to be a priori evident that this need be the case. In fact, would it not be more reasonable to assume a set of simpler and more self-evident axioms about probability and then attempt to prove that such a constant limiting frequency does in some sense exist? The latter approach is the modern axiomatic approach to probability theory that we shall adopt in this text. In particular, we shall assume that, for each event E in the sample space S, there exists a value P(E), referred to as the probability of E. We shall then assume that all these probabilities satisfy a certain set of axioms, which, we hope the reader will agree, is in accordance with our intuitive notion of probability.

Consider an experiment whose sample space is S. For each event E of the sample space S, we assume that a number P(E) is defined and satisfies the following three axioms:

> The three axioms of probability
>
> **Axiom 1**
>
> $0\le P(E)\le 1$
>
> **Axiom 2**
>
> $P(S)=1$
>
> **Axiom 3**
>
> For any sequence of mutually exclusive events E1, E2, ... (that is, events for which $E_i E_j=\emptyset$ when $i\neq j$),
>
> $P(\cup_{i=1}^{\infty} E_i)= \sum_{i=1}^{\infty} P(E_i)$
>
> We refer to P(E) as the *probability* of the event E.

Thus, **Axiom 1** states that the probability that the outcome of the experiment is an outcome in E is some

number between 0 and 1. **Axiom 2** states that, with probability 1, the outcome will be a point in the sample space S. **Axiom 3** states that, for any sequence of mutually exclusive events, the probability of at least one of these events occurring is just the sum of their respective probabilities.

If we consider a sequence of events E 1 , E 2 , . . . , , where E 1 =S and E 1 =∅ for i>1, then, because the events are mutually exclusive and because S=∪i=1∞Ei, we have, from **Axiom 3**,

P(S)=∑i=1∞P(Ei)=P(S)+∑i=2∞P(Ø)

implying that

P(Ø)=0

That is, the null event has probability 0 of occurring.

Note that it follows that, for any finite sequence of mutually exclusive events E1, E2, . . ., En,

P(∪ 1 n E i)= Σ i=1 n P(E i) (3.1)

This equation follows from **Axiom 3** by defining Ei as the null event for all values of i greater than n. **Axiom 3** is equivalent to **Equation (3.1)** when the sample space is finite. (Why?) However, the added generality of **Axiom 3** is necessary when the sample space consists of an infinite number of points.

Example 3a

If our experiment consists of tossing a coin and if we assume that a head is as likely to appear as a tail, then we would have

P({H})=P({T})= 1 2

On the other hand, if the coin were biased and we believed that a head were twice as likely to appear as a tail, then we would have

P({H})=23 P({T})=13

Example 3b

If a die is rolled and we suppose that all six sides are equally likely to appear, then we would have P({1})=P({2})=P({3})=P({4})=P({5})=P({6})=16. From **Axiom 3**, it would thus follow that the probability of rolling an even number would equal

P({2,4,6})=P({2})+P({4})+P({6})=12

The assumption of the existence of a set function P, defined on the events of a sample space S and satisfying **Axioms 1**, **2**, and **3**, constitutes the modern mathematical approach to probability theory. It is hoped that the reader will agree that the axioms are natural and in accordance with our intuitive concept of probability as related to chance and randomness. Furthermore, using these axioms, we shall be able to prove that if an experiment is repeated over and over again, then, with probability 1, the proportion of time during which any specific event *E* occurs will equal P(E). This result, known as the strong law of large numbers, is presented in **Chapter 8**. In addition, we present another possible interpretation of probability– as being a measure of belief–in **Section 2.7**.

Technical Remark. We have supposed that P(E) is defined for all the events E of the sample space. Actually, when the sample space is an uncountably infinite set, P(E) is defined only for a class of events called *measurable*. However, this restriction need not concern us, as all events of any practical interest are

measurable.

2.4 Some Simple Propositions

In this section, we prove some simple propositions regarding probabilities. We first note that since E and Ec are always mutually exclusive and since E∪Ec=S, we have, by **Axioms 2** and **3**,

$$1=P(S)=P(E\cup Ec)=P(E)+P(Ec)$$

Or, equivalently, we have **Proposition 4.1**.

$$P(Ec)=1-P(E)$$

Proposition 4.1

In words, **Proposition 4.1** states that the probability that an event does not occur is 1 minus the probability that it does occur. For instance, if the probability of obtaining a head on the toss of a coin is 38, then the probability of obtaining a tail must be 5 8 .

Our second proposition states that if the event E is contained in the event F, then the probability of E is no greater than the probability of F.

Proposition 4.2

If E⊂F, then P(E)≤P(F).

Proof. Since E⊂F, it follows that we can express F as

$$F=E\cup EcF$$

Hence, because E and EcF are mutually exclusive, we obtain, from **Axiom 3**,

$$P(F)=P(E)+P(E c F)$$

which proves the result, since P(EcF)≥0.

Proposition 4.2 tells us, for instance, that the probability of rolling a 1 with a die is less than or equal to the probability of rolling an odd value with the die.

The next proposition gives the relationship between the probability of the union of two events, expressed in terms of the individual probabilities, and the probability of the intersection of the events.

$$P(E\cup F)=P(E)+P(F)-P(EF)$$

Proposition 4.3

Proof To derive a formula for P(E∪F), we first note that E∪F can be written as the union of the two disjoint events E and EcF. Thus, from **Axiom 3**, we obtain

$$P(E\cup F) =P(E\cup E c F) =P(E)+P(E c F)$$

Furthermore, since F=EF∪EcF, we again obtain from **Axiom 3**

$$P(F)=P(EF)+P(EcF)$$

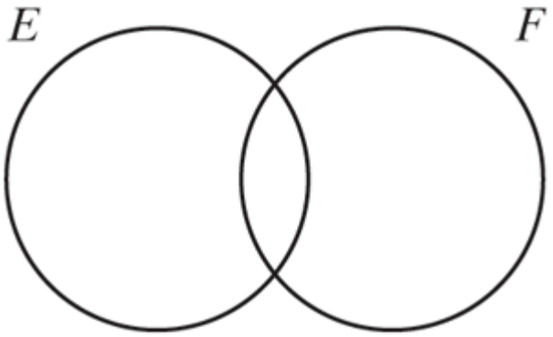
or, equivalently,

$$P(EcF)=P(F)-P(EF)$$

thereby completing the proof.

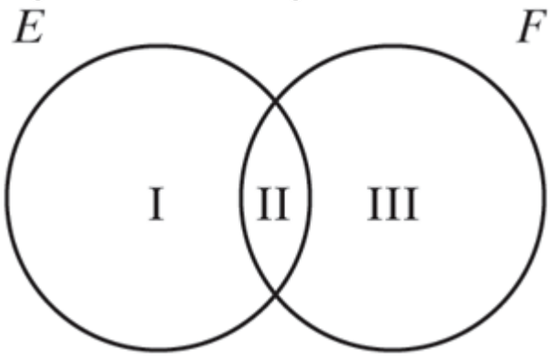
Proposition 4.3 could also have been proved by making use of the Venn diagram in **Figure 2.4**.

Figure 2.4: Venn diagram.



Let us divide E∪F into three mutually exclusive sections, as shown in **Figure 2.5**. In words, section I represents all the points in E that are not in F (that is, EFc), section II represents all points both in E and in F (that is, *EF*), and section III represents all points in F that are not in E (that is, EcF).

Figure 2.5 Venn diagram in sections.



From **Figure 2.5**, we see that

$$E\cup F = I\cup II\cup III \quad E = I\cup II \quad F = II\cup III$$

As I, II, and III are mutually exclusive, it follows from **Axiom 3** that

$$P(E\cup F) = P(I)+P(II)+P(III) \quad P(E) = P(I)+P(II) \quad P(F) = P(II)+P(III)$$

which shows that

$$P(E\cup F)=P(E)+P(F)-P(II)$$

and **Proposition 4.3** is proved, since II = *EF*.

Example 4a

J is taking two books along on her holiday vacation. With probability .5, she will like the first book; with probability .4, she will like the second book; and with probability .3, she will like both books. What is the

probability that she likes neither book?

Solution

Let Bi denote the event that J likes book i, i=1, 2. Then the probability that she likes at least one of the books is

P(B1∪B2)=P(B1)+P(B2)−P(B1B2)=.5+.4−.3=.6

Because the event that J likes neither book is the complement of the event that she likes at least one of them, we obtain the result

P(B1cB2c)=P((B1∪B2)c)=1−P(B1∪B2)=.4

We may also calculate the probability that any one of the three events E, F, and G occurs, namely,

P(E∪F∪G)=P[(E∪F)∪G]

which, by **Proposition 4.3**, equals

P(E∪F)+P(G)−P[(E∪F)G]

Now, it follows from the distributive law that the events (E∪F)G and EG∪FG are equivalent; hence, from the preceding equations, we obtain

P(E∪F∪G) =P(E)+P(F)−P(EF)+P(G)−P(EG∪FG) =P(E)+P(F)−P(EF)+P(G)−P(EG)−P(FG)+P(EGFG) =P(E)+P(F)+P(G)+P(EF)−P(EG)−P(FG)+P(EFG)

In fact, the following proposition, known as the *inclusion–exclusion identity*, can be proved by mathematical induction:

Proposition 4.4

P(E 1 ∪ E 2 ∪⋯∪ E n)= ∑ i=1 n P(E i)− ∑ i 1 < i 2 P(E i 1 E i 2)+⋯ + (−1) r+1 ∑ i 1 < i 2 <⋯< i r P(E i 1 E i 2 ⋯ E i r) +⋯+ (−1) n+1 P(E 1 E 2 ⋯ E n)

The summation ∑i1<i2<⋯<irP(Ei1Ei2⋯Eir) is taken over all of the (nr) possible subsets of size r of the set {1, 2 … ,n}.

In words, **Proposition 4.4** states that the probability of the union of n events equals the sum of the probabilities of these events taken one at a time, minus the sum of the probabilities of these events taken two at a time, plus the sum of the probabilities of these events taken three at a time, and so on.

Remarks 1. For a noninductive argument for **Proposition 4.4**, note first that if an outcome of the sample space is not a member of any of the sets Ei, then its probability does not contribute anything to either side of the equality. Now, suppose that an outcome is in exactly m of the events Ei, where m>0. Then, since it is in ∪iEi, its probability is counted once in P(∪iEi) ; also, as this outcome is contained in (mk) subsets of the type Ei1Ei2⋯Eik, its probability is counted

(m1)−(m2)+(m3)−⋯±(mm)

times on the right of the equality sign in **Proposition 4.4**. Thus, for m>0, we must show that

1=(m 1)−(m 2)+(m 3)− ⋯ ±(m m)

However, since 1=(m0), the preceding equation is equivalent to

$$\sum i=0\ m\ (\ m\ i\ )\ (-1)\ i\ =0$$

and the latter equation follows from the binomial theorem, since

$$0=(-1+1)m=\sum i=0m(mi)(-1)i(1)m-i$$

2. The following is a succinct way of writing the inclusion–exclusion identity:
   $P(\cup i=1nEi)=\sum r=1n(-1)r+1\sum i1<\cdots<irP(Ei1\cdots Eir)$

3. In the inclusion–exclusion identity, going out one term results in an upper bound on the probability of the union, going out two terms results in a lower bound on the probability, going out three terms results in an upper bound on the probability, going out four terms results in a lower bound, and so on. That is, for events E1,...,En, we have
   $$P(\ \cup\ i=1\ n\ E\ i\ )\leq\ \sum\ i=1\ n\ P(\ E\ i\ ) \tag{4.1}$$

   $$P(\cup i=1nEi)\geq\sum i=1nP(Ei)-\sum j<iP(EiEj) \tag{4.2}$$

   $$P(\cup i=1nEi)\leq\sum i=1nP(Ei)-\sum j<iP(EiEj)+\sum k<j<iP(EiEjEk) \tag{4.3}$$

and so on. To prove the validity of these bounds, note the identity

$$\cup i=1nEi=E1\cup E1cE2\cup E1cE2cE3\cup\cdots\cup E1c\cdots En-1cEn$$

That is, at least one of the events Ei occurs if E1 occurs, or if E1 does not occur but E2 does, or if E1 and E2 do not occur but E3 does, and so on. Because the right-hand side is the union of disjoint events, we obtain

$$P(\ \cup\ i=1\ n\ E\ i\ )\ =P(\ E\ 1\ )+P(\ E\ 1\ c\ E\ 2\ )+P(\ E\ 1\ c\ E\ 2\ c\ E\ 3\ )+...+P(\ E\ 1\ c\ \cdots\ E\ n-1\ c\ E\ n\ )\ =P(\ E\ i\ )+\sum\ i=2\ n\ P(\ E\ 1\ c\ \cdots\ E\ i-1\ c\ E\ i\ ) \tag{4.4}$$

Now, let $B\ i\ =\ E\ 1\ c\ \cdots\ E\ i-1\ c\ =\ (\ \cup\ j\ <i\ E\ j\ )\ c$ be the event that none of the first i−1 events occurs. Applying the identity

$$P(Ei)=P(BiEi)+P(BicEi)$$

shows that

$$P(\ E\ i\ )=P(\ E\ 1\ c\ \ldots\ E\ i-1\ c\ E\ i\ )=P(\ E\ i\ )+P(\ \cup\ j<i\ E\ i\ E\ j\ )$$

or, equivalently,

$$P(E1c\cdots Ei-1cEi)=P(Ei)-P(\cup j<iEiEj)$$

Substituting this equation into **(4.4)** yields

$$P(\cup i=1nEi)=\sum iP(Ei)-\sum iP(\cup j<iEiEj) \tag{4.5}$$

Because probabilities are always nonnegative, Inequality (4.1) follows directly from **Equation (4.5)**. Now, fixing i and applying Inequality (1) to $P(\cup j<iEiEj)$ yields

$$P(\cup j<iEiEj)\leq\sum j<iP(EiEj)$$

which, by **Equation (4.5)**, gives Inequality (4.2). Similarly, fixing *i* and applying Inequality (4.2) to $P(\cup_{j<i} E_i E_j)$ yields

$$P(\cup_{j<i} E_i E_j) \geq \sum_{j<i} P(E_i E_j) - \sum_{k<j<i} P(E_i E_j E_i E_k) = \sum_{j<i} P(E_i E_j) - \sum_{k<j<i} P(E_i E_j E_k)$$

which, by **Equation (4.5)**, gives Inequality (4.3). The next inclusion–exclusion inequality is now obtained by fixing i and applying Inequality (4.3) to $P(\cup_{j<i} E_i E_j)$, and so on.

The first inclusion-exclusion inequality, namely that

$$P(\cup_{i=1}^{n} E_i) \leq \sum_{i=1}^{n} P(E_i)$$

is known as *Boole's inequality*.

2.5 Sample Spaces Having Equally Likely Outcomes

In many experiments, it is natural to assume that all outcomes in the sample space are equally likely to occur. That is, consider an experiment whose sample space S is a finite set, say, S={1, 2, ... , N}. Then, it is often natural to assume that

$$P(\{1\})=P(\{2\})=\cdots=P(\{N\})$$

which implies, from **Axioms 2** and **3** (why?), that

$$P(\{i\})=1N \quad i=1, 2, \ldots , N$$

From this equation, it follows from **Axiom 3** that, for any event E,

$$P(E)= \text{number of outcomes in E number of outcomes in S}$$

In words, if we assume that all outcomes of an experiment are equally likely to occur, then the probability of any event E equals the proportion of outcomes in the sample space that are contained in E.

Example 5a

If two dice are rolled, what is the probability that the sum of the upturned faces will equal 7?

Solution

We shall solve this problem under the assumption that all of the 36 possible outcomes are equally likely. Since there are 6 possible outcomes–namely, (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), and (6, 1)–that result in the sum of the dice being equal to 7, the desired probability is 636=16.

Example 5b

If 3 balls are “randomly drawn” from a bowl containing 6 white and 5 black balls, what is the probability that one of the balls is white and the other two black?

Solution

If we regard the balls as being distinguishable and the order in which they are selected as being relevant, then the sample space consists of $11\cdot10\cdot9=990$ outcomes. Furthermore, there are $6\cdot5\cdot4=120$ outcomes in which the first ball selected is white and the other two are black; $5\cdot6\cdot4=120$ outcomes in which the first is black, the second is white, and the third is black; and $5\cdot4\cdot6=120$ in which the first two are black and the third is white. Hence, assuming that “randomly drawn” means that each outcome in the sample space is equally likely to occur, we see that the desired probability is

$$120+120+120990=411$$

This problem could also have been solved by regarding the outcome of the experiment as the unordered set of drawn balls. From this point of view, there are $(113)=165$ outcomes in the sample space. Now, each set of 3 balls corresponds to 3! outcomes when the order of selection is noted. As a result, if all outcomes are assumed equally likely when the order of selection is noted, then it follows that they remain equally likely when the outcome is taken to be the unordered set of selected balls. Hence, using the latter representation of the experiment, we see that the desired probability is

$$(61)\,(52)(113)=411$$

which, of course, agrees with the answer obtained previously.

When the experiment consists of a random selection of k items from a set of n items, we have the flexibility of either letting the outcome of the experiment be the ordered selection of the k items or letting it be the unordered set of items selected. In the former case, we would assume that each new selection is equally likely to be any of the so far unselected items of the set, and in the latter case, we would assume that all (nk) possible subsets of k items are equally likely to be the set selected. For instance, suppose 5 people are to be randomly selected from a group of 20 individuals consisting of 10 married couples, and we want to determine P(N), the probability that the 5 chosen are all unrelated. (That is, no two are married to each other.) If we regard the sample space as the set of 5 people chosen, then there are (20 5) equally likely outcomes. An outcome that does not contain a married couple can be thought of as being the result of a six-stage experiment: In the first stage, 5 of the 10 couples to have a member in the group are chosen; in the next 5 stages, 1 of the 2 members of each of these couples is selected. Thus, there are (105)25 possible outcomes in which the 5 members selected are unrelated, yielding the desired probability of

P(N)=(105)25(205)

In contrast, we could let the outcome of the experiment be the *ordered* selection of the 5 individuals. In this setting, there are 20·19·18·17·16 equally likely outcomes, of which 20·18·16·14·12 outcomes result in a group of 5 unrelated individuals, yielding the result

P(N)=20·18·16·14·1220·19·18·17·16

We leave it for the reader to verify that the two answers are identical.

Example 5c

A committee of 5 is to be selected from a group of 6 men and 9 women. If the selection is made randomly, what is the probability that the committee consists of 3 men and 2 women?

Solution

Because each of the (15 5) possible committees is equally likely to be selected, the desired probability is

(63)(92)(155)=2401001

Example 5d

An urn contains n balls, one of which is special. If k of these balls are withdrawn one at a time, with each selection being equally likely to be any of the balls that remain at the time, what is the probability that the special ball is chosen?

Solution

Since all of the balls are treated in an identical manner, it follows that the set of k balls selected is equally likely to be any of the (nk) sets of k balls. Therefore,

P{special ball is selected}=(11)(n−1k−1)(nk)=kn

We could also have obtained this result by letting Ai denote the event that the special ball is the ith ball to be chosen, i=1, ... , k. Then, since each one of the n balls is equally likely to be the ith ball chosen, it follows that P(Ai)=1/n. Hence, because these events are clearly mutually exclusive, we have

P{special ball is selected}=P(∪ i=1 k A i)= ∑ i=1 k P(A i)= k n

We could also have argued that P(A i)=1/n, by noting that there are n(n−1)⋯(n−k+1)=n!/(n−k)! equally likely outcomes of the experiment, of which (n−1)(n−2)⋯(n−i+1)(1)(n−i)⋯(n−k+1)=(n−1)!/(n−k)! result in the special ball being the ith one chosen. From this reasoning, it follows that

P(Ai)=(n−1)!n!=1n

Example 5e

Suppose that n+m balls, of which n are red and m are blue, are arranged in a linear order in such a way that all (n+m)! possible orderings are equally likely. If we record the result of this experiment by listing only the colors of the successive balls, show that all the possible results remain equally likely.

Solution

Consider any one of the (n+m)! possible orderings, and note that any permutation of the red balls among themselves and of the blue balls among themselves does not change the sequence of colors. As a result, every ordering of colorings corresponds to n! m! different orderings of the n+m balls, so every ordering of the colors has probability n!m! (n+m)! of occurring.

For example, suppose that there are 2 red balls, numbered r1,r2, and 2 blue balls, numbered b1,b2. Then, of the 4! possible orderings, there will be 2! 2! orderings that result in any specified color combination. For instance, the following orderings result in the successive balls alternating in color, with a red ball first:

r 1 , b 1 , r 2 , b 2 r 1 , b 2 , r 2 , b 1 r 2 , b 1 , r 1 , b 2 r 2 , b 2 , r 1 , b 1

Therefore, each of the possible orderings of the colors has probability 424=16 of occurring.

Example 5f

A poker hand consists of 5 cards. If the cards have distinct consecutive values and are not all of the same suit, we say that the hand is a straight. For instance, a hand consisting of the five of spades, six of spades, seven of spades, eight of spades, and nine of hearts is a straight. What is the probability that one is dealt a straight?

Solution

We start by assuming that all (525) possible poker hands are equally likely. To determine the number of outcomes that are straights, let us first determine the number of possible outcomes for which the poker hand consists of an ace, two, three, four, and five (the suits being irrelevant). Since the ace can be any 1 of the 4 possible aces, and similarly for the two, three, four, and five, it follows that there are 45 outcomes leading to exactly one ace, two, three, four, and five. Hence, since in 4 of these outcomes all the cards will be of the same suit (such a hand is called a straight flush), it follows that there are 45−4 hands that make up a straight of the form ace, two, three, four, and five. Similarly, there are 45−4 hands that make up a straight of the form ten, jack, queen, king, and ace. Thus, there are 10(45−4) hands that are straights, and it follows that the desired probability is

10(45−4)(525)≈.0039

Example 5g

A 5-card poker hand is said to be a full house if it consists of 3 cards of the same denomination and 2 other cards of the same denomination (of course, different from the first denomination). Thus, a full house is three of a kind plus a pair. What is the probability that one is dealt a full house?

Solution

Again, we assume that all (525) possible hands are equally likely. To determine the number of possible full houses, we first note that there are (42)(43) different combinations of, say, 2 tens and 3 jacks. Because there are 13 different choices for the kind of pair and, after a pair has been chosen, there are 12 other choices for the denomination of the remaining 3 cards, it follows that the probability of a full house is

$13 \cdot 12 \cdot (42)(43)(525) \approx .0014$

Example 5h

In the game of bridge, the entire deck of 52 cards is dealt out to 4 players. What is the probability that

a. one of the players receives all 13 spades;
b. each player receives 1 ace?

Solution

a. Letting Ei be the event that hand i has all 13 spades, then
$P(E_i) = 1 / \binom{52}{13}, \quad i=1,2,3,4$

   Because the events Ei, i=1, 2, 3, 4, are mutually exclusive, the probability that one of the hands is dealt all 13 spades is
$P(\cup_{i=1}^{4} E_i) = \sum_{i=1}^{4} P(E_i) = 4/\binom{52}{13} \approx 6.3 \times 10^{-12}$

b. Let the outcome of the experiment be the sets of 13 cards of each of the players 1, 2, 3, 4. To determine the number of outcomes in which each of the distinct players receives exactly 1 ace, put aside the aces and note that there are (4812, 12, 12, 12) possible divisions of the other 48 cards when each player is to receive 12. Because there are 4! ways of dividing the 4 aces so that each player receives 1, we see that the number of possible outcomes in which each player receives exactly 1 ace is 4!(4812, 12, 12, 12).

As there are $\binom{52}{13, 13, 13, 13}$ possible hands, the desired probability is thus

$4!(4812,12,12,12)/(5213,13,13,13) \approx .1055$

Some results in probability are quite surprising when initially encountered. Our next two examples illustrate this phenomenon.

Example 5i

If n people are present in a room, what is the probability that no two of them celebrate their birthday on the same day of the year? How large need n be so that this probability is less than 12?

Solution

As each person can celebrate his or her birthday on any one of 365 days, there are a total of $(365)^n$ possible outcomes. (We are ignoring the possibility of someone having been born on February 29.) Assuming that each outcome is equally likely, we see that the desired probability is $(365)(364)(363) \cdots (365-n+1)/(365)^n$. It is a rather surprising fact that when $n \geq 23$, this probability is less than 12. That is, if there are 23 or more people in a room, then the probability that at least two of them have the same birthday exceeds 12. Many people are initially surprised by this result, since 23 seems so small in relation to 365, the number of days of the year. However, every pair of individuals has probability 365 (365) 2 = 1 365 of having the same birthday, and in a group of 23 people, there are

(232)=253 different pairs of individuals. Looked at this way, the result no longer seems so surprising.

When there are 50 people in the room, the probability that at least two share the same birthday is approximately .970, and with 100 persons in the room, the odds are better than 3,000,000:1. (That is, the probability is greater than 3×1063×106+1 that at least two people have the same birthday.)

Example 5j

A deck of 52 playing cards is shuffled, and the cards are turned up one at a time until the first ace appears. Is the next card–that is, the card following the first ace–more likely to be the ace of spades or the two of clubs?

Solution

To determine the probability that the card following the first ace is the ace of spades, we need to calculate how many of the (52)! possible orderings of the cards have the ace of spades immediately following the first ace. To begin, note that each ordering of the 52 cards can be obtained by first ordering the 51 cards different from the ace of spades and then inserting the ace of spades into that ordering. Furthermore, for each of the (51)! orderings of the other cards, there is only one place where the ace of spades can be placed so that it follows the first ace. For instance, if the ordering of the other 51 cards is

4c, 6h, Jd, 5s, Ac, 7d, ... , Kh

then the only insertion of the ace of spades into this ordering that results in its following the first ace is

4c, 6h, Jd, 5s, Ac, As, 7d, ... , Kh

Therefore, there are (51)! orderings that result in the ace of spades following the first ace, so

P{the ace of spades follows the first ace}=(51)!(52)!=152

In fact, by exactly the same argument, it follows that the probability that the two of clubs (or any other specified card) follows the first ace is also 152. In other words, each of the 52 cards of the deck is equally likely to be the one that follows the first ace!

Many people find this result rather surprising. Indeed, a common reaction is to suppose initially that it is more likely that the two of clubs (rather than the ace of spades) follows the first ace, since that first ace might itself be the ace of spades. This reaction is often followed by the realization that the two of clubs might itself appear before the first ace, thus negating its chance of immediately following the first ace. However, as there is one chance in four that the ace of spades will be the first ace (because all 4 aces are equally likely to be first) and only one chance in five that the two of clubs will appear before the first ace (because each of the set of 5 cards consisting of the two of clubs and the 4 aces is equally likely to be the first of this set to appear), it again appears that the two of clubs is more likely. However, this is not the case, and our more complete analysis shows that they are equally likely.

Example 5k

A football team consists of 20 offensive and 20 defensive players. The players are to be paired in groups of 2 for the purpose of determining roommates. If the pairing is done at random, what is the probability that there are no offensive–defensive roommate pairs? What is the probability that there are 2i offensive–defensive roommate pairs, i=1, 2, ... , 10?

Solution

There are

(402, 2, ... , 2)=(40)!(2!)20

ways of dividing the 40 players into 20 *ordered* pairs of two each. (That is, there are (40)!/ 2 20 ways of dividing the players into a *first* pair, a *second* pair, and so on.) Hence, there are (40)!/220(20)! ways of dividing the players into (unordered) pairs of 2 each. Furthermore, since a division will result in no offensive–defensive pairs if the offensive (and defensive) players are paired among themselves, it follows that there are [(20)!/210(10)!]2 such divisions. Hence, the probability of no offensive–defensive roommate pairs, call it P0, is given by

P0=((20)!210(10)!)2(40)!220(20)!=[(20)!]3[(10)!]2(40)!

To determine P2i, the probability that there are 2i offensive–defensive pairs, we first note that there are (202i)2 ways of selecting the 2i offensive players and the 2i defensive players who are to be in the offensive–defensive pairs. These 4i players can then be paired up into (2i)! possible offensive–defensive pairs. (This is so because the first offensive player can be paired with any of the 2i defensive players, the second offensive player with any of the remaining 2i−1 defensive players, and so on.) As the remaining 20−2i offensive (and defensive) players must be paired among themselves, it follows that there are

(202i)2(2i)![(20−2i)!210−i(10−i)!]2

divisions that lead to 2i offensive–defensive pairs. Hence,

P2i=(202i)2(2i)![(20−2i)!210−i(10−i)!]2(40)!220(20)! i=0, 1, ... , 10

The P2i, i=0, 1, ..., 10, can now be computed, or they can be approximated by making use of a result of Stirling, which shows that n! can be approximated by nn+1/2e−n2π. For instance, we obtain

P 0 ≈1.3403× 10 −6 P 10 ≈.345861 P 20 ≈7.6068× 10 −6

Our next three examples illustrate the usefulness of the inclusion–exclusion identity (**Proposition 4.4**). In **Example 51**, the introduction of probability enables us to obtain a quick solution to a counting problem.

Example 5l

A total of 36 members of a club play tennis, 28 play squash, and 18 play badminton. Furthermore, 22 of the members play both tennis and squash, 12 play both tennis and badminton, 9 play both squash and badminton, and 4 play all three sports. How many members of this club play at least one of three sports?

Solution

Let N denote the number of members of the club, and introduce probability by assuming that a member of the club is randomly selected. If, for any subset C of members of the club, we let P(C) denote the probability that the selected member is contained in C, then

P(C)=number of members in CN

Now, with T being the set of members that plays tennis, S being the set that plays squash, and B being the set that plays badminton, we have, from **Proposition 4.4**,

P(T∪S∪B) =P(T)+P(S)+P(B)−P(TS)−P(TB)−P(SB)+P(TSB) = 36+28+18−22−12−9+4 N = 43 N

Hence, we can conclude that 43 members play at least one of the sports.

The next example in this section not only possesses the virtue of giving rise to a somewhat surprising answer, but is also of theoretical interest.

Example 5m The Matching Problem

Suppose that each of N men at a party throws his hat into the center of the room. The hats are first mixed up, and then each man randomly selects a hat. What is the probability that none of the men selects his own hat?

Solution

We first calculate the complementary probability of at least one man selecting his own hat. Let us denote by Ei, i=1, 2, ... , N the event that the ith man selects his own hat. Now, by the inclusion-exclusion identity P(∪i=1NEi), the probability that at least one of the men selects his own hat, is given by

$$P(\cup i=1 N E i)= \sum i=1 N P(E i)- \sum i 1 < i 2 P(E i 1 E i 2)+\cdots + (-1) n+1 \sum i 1 < i 2 \cdots< i n P(E i 1 E i 2 \cdots E i n) +\cdots+ (-1) N+1 P(E 1 E 2 \cdots E N)$$

If we regard the outcome of this experiment as a vector of *N* numbers, where the ith element is the number of the hat drawn by the ith man, then there are N! possible outcomes. [The outcome (1, 2, 3, ... , N) means, for example, that each man selects his own hat.] Furthermore, Ei1Ei2...Ein, the event that each of the *n* men i1, i2, ..., in selects his own hat, can occur in any of $(N-n)(N-n-1)\cdots 3\cdot 2\cdot 1=(N-n)!$ possible ways; for, of the remaining N−n men, the first can select any of N−n hats, the second can then select any of N−n−1 hats, and so on. Hence, assuming that all N! possible outcomes are equally likely, we see that

$$P(Ei1Ei2\cdots Ein)=(N-n)!N!$$

Also, as there are (Nn) terms in $\sum i1<i2\cdots<inP(Ei1Ei2\cdots Ein)$, it follows that

$$\sum i1<i2\cdots<inP(Ei1Ei2\cdots Ein)=N!(N-n)!n!(N-n)!N!=1n!$$

Thus,

$$P(\cup i=1NEi)=1-12!+13!-\cdots+(-1)N+11N!$$

Hence, the probability that none of the men selects his own hat is

$$1-1+12!-13!+...+(-1)NN!=\sum i=0N(-1)i/i!$$

Upon letting x=−1 in the identity $e x = \sum i=0 \infty x i /i!$, the preceding probability when N is large is seen to be approximately equal to $e-1\approx.3679$. In other words, for N large, the probability that none of the men selects his own hat is approximately .37. (How many readers would have incorrectly thought that this probability would go to 1 as $N\rightarrow\infty$?)

For another illustration of the usefulness of the inclusion-exclusion identity, consider the following example.

Example 5n

Compute the probability that if 10 married couples are seated at random at a round table, then no wife sits next to her husband.

Solution

If we let Ei, i=1, 2, ... , 10 denote the event that the ith couple sit next to each other, it follows that the desired probability is 1−P(∪i=110Ei). Now, from the inclusion-exclusion identity,

P(∪ 1 10 E i)= ∑ 1 10 P(E i)−⋯+ (−1) n+1 ∑ i 1 < i 2 <⋯< i n P(E i 1 E i 2 ⋯ E i n) +⋯−P(E 1 E 2 ⋯ E 10)

To compute P(Ei1Ei2⋯Ein), we first note that there are 19! ways of arranging 20 people around a round table. (Why?) The number of arrangements that result in a specified set of n men sitting next to their wives can most easily be obtained by first thinking of each of the n married couples as being single entities. If this were the case, then we would need to arrange 20−n entities around a round table, and there are clearly (20−n−1)! such arrangements. Finally, since each of the n married couples can be arranged next to each other in one of two possible ways, it follows that there are 2n(20−n−1)! arrangements that result in a specified set of n men each sitting next to their wives. Therefore,

P(Ei1Ei2⋯Ein)=2n(19−n)!(19)!

Thus, from **Proposition 4.4**, we obtain that the probability that at least one married couple sits together is

(101)21(18)!(19)!−(102)22(17)!(19)!+(103)23(16)!(19)!−⋯−(1010)2109!(19)!≈.6605

and the desired probability is approximately .3395.

***Example 5o Runs**

Consider an athletic team that had just finished its season with a final record of n wins and m losses. By examining the sequence of wins and losses, we are hoping to determine whether the team had stretches of games in which it was more likely to win than at other times. One way to gain some insight into this question is to count the number of runs of wins and then see how likely that result would be when all (n+m)!/(n! m!) orderings of the n wins and m losses are assumed equally likely. By a run of wins, we mean a consecutive sequence of wins. For instance, if n=10, m=6, and the sequence of outcomes was *WWLLWWWLWLLLWWWW*, then there would be 4 runs of wins–the first run being of size 2, the second of size 3, the third of size 1, and the fourth of size 4.

Suppose now that a team has n wins and m losses. Assuming that all (n+m)!/ (n! m!)=(n+mn) orderings are equally likely, let us determine the probability that there will be exactly r runs of wins. To do so, consider first any vector of positive integers x 1 , x 2 ,…, x r with x1+⋯+xr=n, and let us see how many outcomes result in *r* runs of wins in which the ith run is of size xi, i=1, ... , r. For any such outcome, if we let y1 denote the number of losses before the first run of wins, y2 the number of losses between the first 2 runs of wins, ... , yr+1 the number of losses after the last run of wins, then the yi satisfy

y1+y2+⋯+yr+1=m y1≥0,yr+1≥0,yi>0,i=2, ... , r

and the outcome can be represented schematically as

LL ... L‿y1 WW ... W‿x1 L ... L‿y2 WW ... W‿x2 ⋯ WW‿xr L ... L‿yr+1

Hence, the number of outcomes that result in r runs of wins the ith of size x i ,i=1, … r− is equal to the number of integers y1 , ... , yr+1 that satisfy the foregoing, or, equivalently, to the number of positive integers

y ‾ 1 = y 1 +1 y ‾ 1 = y i , i=2,…,r, y ‾ r+1 = y r+1 +1

that satisfy

$$\bar{y}_1+\bar{y}_2+\cdots+\bar{y}_{r+1}=m+2$$

By **Proposition 6.1** in **Chapter 1**, there are $\binom{m+1}{r}$ such outcomes. Hence, the total number of outcomes that result in r runs of wins is $\binom{m+1}{r}$ multiplied by the number of positive integral solutions of $x_1+\cdots+x_r=n$. Thus, again from **Proposition 6.1**, there are $\binom{n-1}{r-1}\binom{m+1}{r}$ outcomes resulting in r runs of wins. As there are $\binom{n+m}{n}$ equally likely outcomes, it follows that

$$P(\{\text{r runs of wins}\})=\frac{\binom{m+1}{r}\binom{n-1}{r-1}}{\binom{m+n}{n}} \quad r\geq 1$$

For example, if n=8 and m=6, then the probability of 7 runs is $\binom{7}{7}\binom{7}{6}/\binom{14}{8}=1/429$ if all $\binom{14}{8}$ outcomes are equally likely. Hence, if the outcome was *WLWLWLWLWWLWLW*, then we might suspect that the team's probability of winning was changing over time. (In particular, the probability that the team wins seems to be quite high when it lost its last game and quite low when it won its last game.) On the other extreme, if the outcome were *WWWWWWWWWLLLLL*, then there would have been only 1 run, and as $P(\{1\text{ run}\})=\binom{7}{1}\binom{7}{0}/\binom{14}{8}=1/429$, it would thus again seem unlikely that the team's probability of winning remained unchanged over its 14 games.